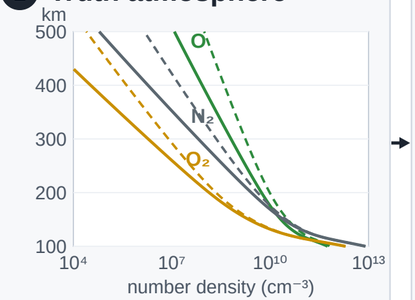
How the performance in Figure 2 was computed

From a model atmosphere to single-unit numbers, one occultation. Each tag below a stage says what kind of input it is:

measured · modelled from data · assumed · method or design choice · stated limit

1 Truth atmosphere

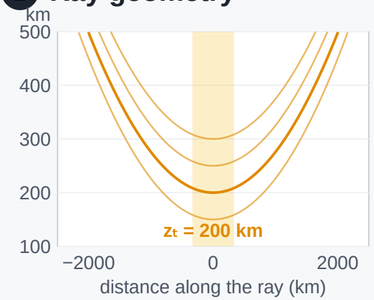


solid: quiet, ap 4 dashed: storm, ap 400

$n_s(z)$, s = O, N_2, O_2
from NRLMSIS 2.1, 2 km grid,
90–800 km

- NRLMSIS stands in for the real
- atmosphere; no flight data
- 21 Mar 2026, dawn, equator
- F10.7 140; ap 4 and ap 400
- helium omitted above 400 km

2 Ray geometry

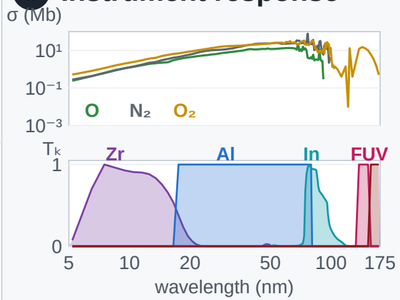


shaded: central 50 % of the column,
±329 km, 5.7° of arc, z_t = 200 km

$N_s(z_t) = \int n_s(r)\, dx$
$r = \sqrt{[(R\oplus + z_t)^2 + x^2]} - R\oplus$
z_t = 110–400 km, every 10 km

- spherical shells: no structure
- along the ray
- path ±3000 km, top 800 km
- tangent sampling every 10 km
- host at 500 km (km per mrad)

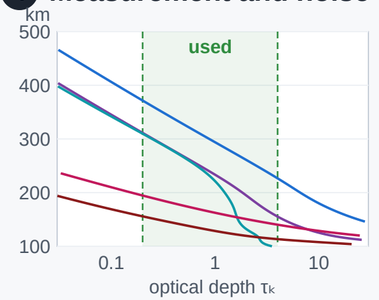
3 Instrument response



$\tau_k(z_t) = -\ln(\, I_k / I_{0k}\,)$
$I_k = \Sigma\lambda\, F\lambda\, T_k(\lambda)\, e^{(-\Sigma_s\, \sigma_s\, N_s)}$
then a Gaussian blur, width w_k

- Zr, In curves: Luxel, digitized
- Al, FUV: rectangular bands
- σ, F: project tables, 0.5–175 nm
- w_k 10.8–23.3 km: slit, diffraction
- no solar-disk brightness profile

4 Measurement and noise

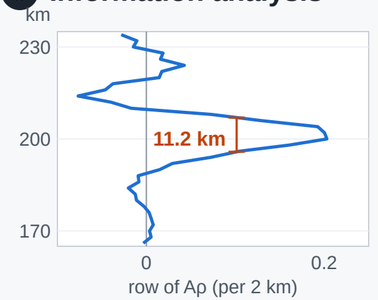


use $0.2 \le \tau_k \le 4$
$\sigma(\tau)^2 = [c_k\, e^{(\tau/2)}]^2 + f^2$
c_k = 1.1–6.1 × 10^{-4}, f = 1 %

- c_k: photon budget, 25 µm slit,
- 4.24 s per 10 km, filter curves
- detector efficiency 0.5
- f = 1 %, correlated over 20 km
- FUV: two-filter stacks

EUV τ_k with noise

5 Information analysis



$x = \ln n_s(z)$, $K = \partial\tau/\partial x$
$\hat{S} = (K^T S_e^{-1} K + S_a^{-1})^{-1}$
$A = \hat{S}\, K^T\, S_e^{-1}\, K$, $\rho = \Sigma_s\, m_s\, n_s$

- linear about the truth
- prior: 0.26 in ln n, 10 km exp.
- width set so the largest error
- over 170–300 km is 10 %
- noise Gaussian, known exactly

6 Readouts → Figure 2

- **P1 Random error** — $\sigma\rho = \sqrt{\text{diag}(M\, \hat{S}\, M^T)}$
- **P1 Covered altitudes** — $\sigma\rho \le 10$ %, $0.8 \le \Sigma_j\, A\rho_{,ij} \le 1.2$
- **P2 Vertical resolution** — width of a row of Aρ = M A P
- **P3 Horizontal** — central 50 % of $\int \rho\, dx$
- **P4 Composition** — $\sqrt{\text{diag}(\hat{S})}$, one species
- **P5 Table** — the above, the checks, and the temperature branch below

- M_i: mass weights $m_s n_s/\rho$
- one setting for every readout
- plots: worst or median per
- 10 km interval
- modelled; no flight data

table row

7 Checks on the analysis



Retrieval: $x_{i+1} = x_a + \hat{S}\, K^T\, S_e^{-1} \times$
$[\, y - F(x_i) + K\,(x_i - x_a)\,]$
Recovers 7 of 8 test atmospheres within
10 %, 170–300 km (not: mixing to 160 km)
+10 % bump, 180–300 km, quiet and storm:
widths within 1 km of stage 5, 8 of 8

- same forward model as the truth, no noise

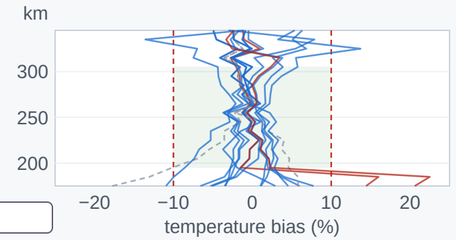
8 Temperature branch

Profile fit
$T(z) = T\infty - (T\infty - 360\ K)$
$\times\, e^{(-s\ (z - 120\ km))}$
Zr, Al, In; 170–400 km

Layer-by-layer peel
$N(z_i) = \Sigma_{j \ge i}\, n_j\, \Delta x_{ij}$
n_s in 10 km shells,
360 km down to 170 km

Hydrostatic temperature
$p(z) = p_{top} + \int \rho\, g\, dz'$
$T = p\ /\ (n\ k_B)$
$p_{top} = n\ k_B\ T_{fit}(355\ km)$

- prior: NRLMSIS; O, N_2 ±45 %; O_2 ±60 % (vs GOLD)
- 30 noisy draws; resolution 10.8 km



within 10 %
195–305 km,
every scored case
red: mixing
to 160 km
dashed: ap 200,
not scored

Not in the model

- out-of-band leak through the filters
- scattered light
- pointing jitter
- smear during a frame (5.2 km at 0.5 frame/s)
- solar-disk brightness profile
- horizontal structure along the ray
- helium above 400 km
- detector nonlinearity
- errors in σ (they enter accuracy, not random error)

Computed with the project simulator: occ.py (forward model), dofs.py (information analysis), retrieve_oe.py (retrieval), hybrid_T.py (temperature);
inputs and results as of 1 October 2026. Stages 1–6 use the Figure 2 atmospheres; stages 7–8 use the test atmospheres of truth.py. P1–P5: Figure 2 panels.